

Decision Tree Pruning Analysis

Introduction

Pruning is a technique used in decision trees to reduce overfitting and improve generalization. In this experiment, we compare the impact of pruning on decision tree performance using two different approaches. The experiment is conducted on the Wine dataset across multiple random states to evaluate the consistency of pruning effects.

Exploratory Data Analysis

An exploratory analysis was performed on the data to gain insights into its structure and characteristics.

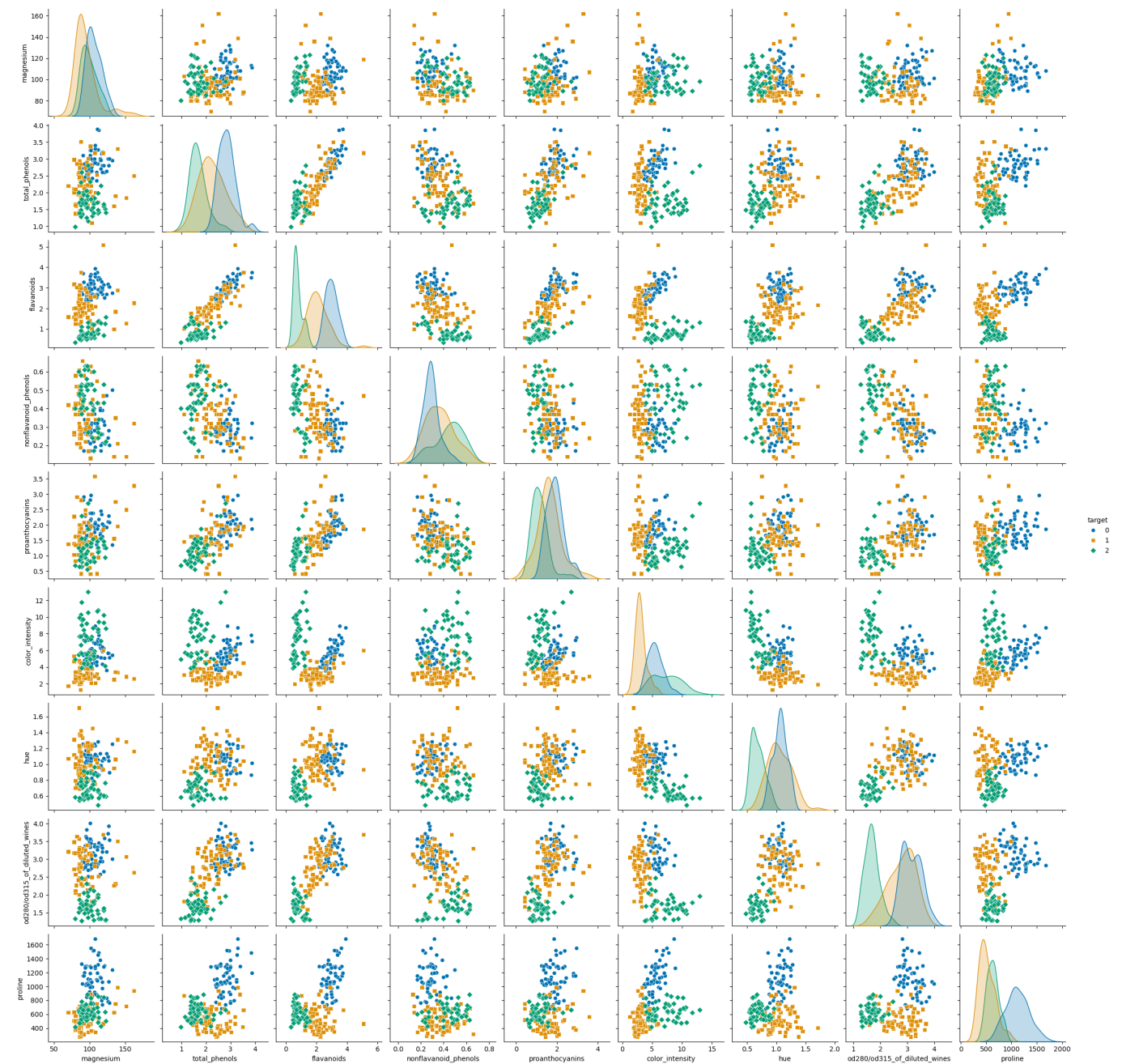
First Five Rows

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	color_intensity	hue	od280/od315_of_diluted_wines	proline
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	0.28	2.29	5.64	1.04	3.92	1065.0
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050.0
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185.0
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480.0
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735.0

Summary Statistics

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	color_intensity	hue	od280/od315_of_diluted_wines	proline
count	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000
mean	13.000618	2.336348	2.366517	19.494944	99.741573	2.295112	2.029270	0.361854	1.590899	5.058090	0.957449	2.611685	
std	0.811827	1.117146	0.274344	3.339564	14.282484	0.625851	0.998859	0.124453	0.572359	2.318286	0.228572	0.709990	
min	11.030000	0.740000	1.360000	10.600000	70.000000	0.980000	0.340000	0.130000	0.410000	1.280000	0.480000	1.270000	
25%	12.362500	1.602500	2.210000	17.200000	88.000000	1.742500	1.205000	0.270000	1.250000	3.220000	0.782500	1.937500	
50%	13.050000	1.865000	2.360000	19.500000	98.000000	2.355000	2.135000	0.340000	1.555000	4.690000	0.965000	2.780000	
75%	13.677500	3.082500	2.557500	21.500000	107.000000	2.800000	2.875000	0.437500	1.950000	6.200000	1.120000	3.170000	
max	14.830000	5.800000	3.230000	30.000000	162.000000	3.880000	5.080000	0.660000	3.580000	13.000000	1.710000	4.000000	

Feature Relationship Visualization



The dataset has many features with good distribution to use but the number of samples makes this task challenging as it is hard to avoid a biased training dataset. This problem will also be observed in the next part showing inefficiency of pruning.

Results

Accuracy Comparison

Random State	Unpruned Accuracy	Pruned Accuracy	Custom Pruned Accuracy (α = 0.05)	Custom Pruned Accuracy (α = 0.1)	Custom Pruned Accuracy (α = 0.2)
1	0.75	0.53	0.53	0.53	0.42
2	0.92	0.75	0.75	0.75	0.75
3	0.78	0.64	0.64	0.64	0.64
4	0.53	0.53	0.53	0.56	0.56

Random State	Unpruned Accuracy	Pruned Accuracy	Custom Pruned Accuracy ($\alpha = 0.05$)	Custom Pruned Accuracy ($\alpha = 0.1$)	Custom Pruned Accuracy ($\alpha = 0.2$)
5	0.83	0.69	0.75	0.75	0.39

Mean Accuracy across Random States:

- **Unpruned:** 0.76
- **Pruned:** 0.63
- **Custom Pruned ($\alpha = 0.05$):** 0.64
- **Custom Pruned ($\alpha = 0.1$):** 0.64
- **Custom Pruned ($\alpha = 0.2$):** 0.55

Discussion

Impact of Pruning on Accuracy

From the results, pruning generally reduces accuracy compared to the unpruned decision tree. The mean test accuracy dropped from **0.76 (unpruned)** to **0.63 (pruned)**, indicating that pruning led to a simpler model but at the cost of performance. This suggests that the original decision trees might not have been excessively overfitting, as the model did not have enough samples in the first place, making aggressive pruning less beneficial.

Reduced Error Pruning vs. Custom Pruning

The **reduced error pruning** method resulted in a mean accuracy of **0.63**, whereas the **custom pruning** method with α values of 0.05 and 0.1 resulted in slightly better accuracy (**0.64**). However, when α was set to 0.2, the accuracy dropped significantly to **0.55**, indicating over-pruning. This suggests that **mild pruning ($\alpha = 0.05, 0.1$) may help reduce complexity without drastically impacting accuracy**, whereas higher α values can remove too many important nodes.

Effect of α on Custom Pruning

- **Lower α values (0.05, 0.1)** resulted in pruning with minimal impact on accuracy, which may suggest that only a few weak branches were removed.
- **Higher α value (0.2)** pruned the tree more aggressively, leading to a significant drop in accuracy (as seen in random states 1 and 5).
- This confirms that **higher α leads to more aggressive pruning**, reducing model complexity but at the cost of performance.

Challenges and Insights

- Despite the stratified splits to avoid biased partitions **Randomness in dataset splits** influenced the pruning impact, as seen in variations across different random states.
- **Choosing the right α value** is crucial—too small has little effect, while too large can severely impact accuracy.
- **The Wine dataset** probably does not suffer from severe overfitting, as pruning did not consistently improve accuracy, meaning other regularization techniques like feature selection or hyperparameter

tuning may be more effective.

Conclusion

The results are honestly worse than I expected. This is probably due to the selected features are not best for this task but feature selection is beyond of this assignment. As mentioned above pruning simplifies decision trees but can negatively impact accuracy if too aggressive. Reduced error pruning and custom pruning with **lower α values (0.05, 0.1)** performed similarly, while **higher α values (0.2)** led to over-pruning.